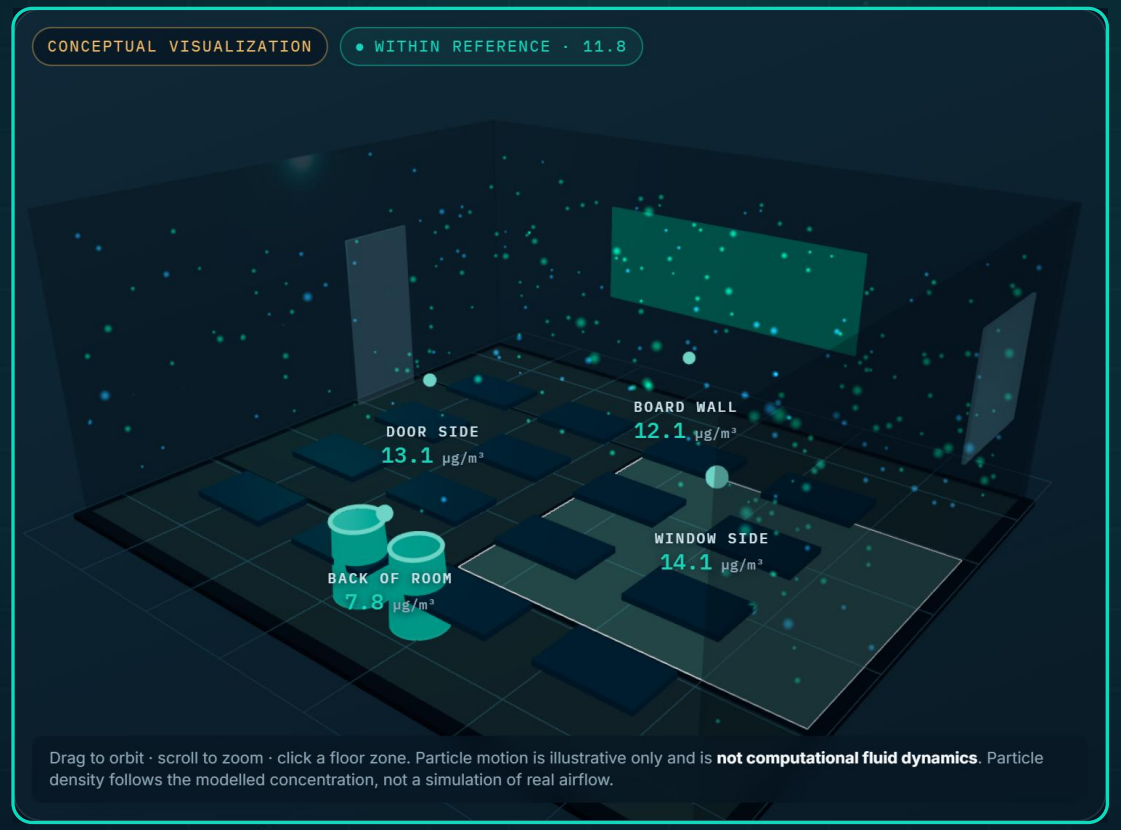
YOUNG PLANET LEADERS · VISION TO ACTION 2026

AIRSHIFT

What if you could see the air you breathe?

Making the Invisible Visible. Turning Awareness into Action.

DETECT → DECIDE → VERIFY → REPLICATE



Screenshot of the working prototype · Air Lab · simulated demonstration data

May Ahmed Mourad · Grade 12 · ADNOC Schools Madinat Zayed, Al Dhafra, UAE · Solo participant

A proposed student-led pilot, subject to school approval. Prototype uses simulated demonstration data.

THE GLOBAL PROBLEM

Almost everyone breathes air above WHO guideline limits.

PROBLEM STATEMENT · PART 1 OF 2

Air pollution is one of the world's most significant environmental health challenges. The World Health Organization reports that almost all of the global population (99%) breathe air that exceeds its guideline limits, and associates ambient and household air pollution with about 7 million premature deaths each year. Fine particles known as PM2.5, smaller than 2.5 micrometres, are a leading concern. But these are global, outdoor statistics. They do not tell a student what is in the air of their own classroom, where they spend much of the day. Most schools have no affordable way to measure indoor air, interpret the readings, choose a fix within budget, or check that it worked.

Outdoor and global ≠ indoor and local. That gap is the problem AIRSHIFT starts with.

99%

of the world's population breathe air that exceeds WHO guideline limits

WHO

~7 million

premature deaths a year associated with ambient and household air pollution combined

WHO

5 µg/m³

WHO 2021 annual guideline for PM2.5. Particles are smaller than 2.5 micrometres

WHO AQG 2021

Sources: WHO, Air pollution (who.int/health-topics/air-pollution); WHO Global air quality guidelines 2021. Global and outdoor figures; they are not measurements of any classroom.

THE LOCAL CHALLENGE

The gap: no school can see its own air.

PROBLEM STATEMENT · PART 2 OF 2

AIRSHIFT begins in Al Dhafra, in the UAE's desert interior. Published monitoring work describes wind-blown desert dust as a major particulate source across Abu Dhabi emirate, but that is an outdoor, regional picture. Whether classroom air at ADNOC Schools Madinat Zayed is affected is simply unknown, because no one has measured it. Indoor particles can come from outdoor dust entering through doors and gaps, foot traffic, chalk, cleaning products and ageing filters. Students, teachers and staff share that air. Our proposed pilot, subject to school approval, starts with a measured baseline instead of assuming a problem.

207 words across both parts (target: about 200)

WHAT WE KNOW · OUTDOOR, REGIONAL

- Desert dust is a major natural particulate source in Abu Dhabi emirate (peer-reviewed monitoring study).
- UAE annual outdoor PM2.5 in 2025: 36.4 µg/m³, 7.3× the WHO annual guideline (IQAir). National, not local.
- In that study PM2.5 was generally under half of PM10, pointing to coarse natural dust rather than fine particles.

WHAT NOBODY KNOWS YET

Indoor PM2.5 in Madinat Zayed classrooms

Not measured. No baseline exists. The pilot measures first, and acts only if the data justifies it.

Sources: Alhebsi et al. 2025, ISPRS Archives XLVIII-4/W17 (doi 10.5194/isprs-archives-XLVIII-4-W17-2025-13-2026); IQAir UAE 2025 (national, outdoor, private provider).

OUR MISSION

See it. Fix it. Prove it. Share it.

To empower schools to understand their air quality, implement affordable evidence-based improvements, and measure their impact. Beginning with a student-led pilot in Al Dhafra, AIRSHIFT aims to create an accessible, replicable model that enables young changemakers worldwide to turn environmental awareness into measurable community action.

01 Baseline

Document PM2.5 in one or two classrooms with dates, windows and sensor caveats.

02 Test

Try one affordable, evidence-based intervention and verify its effect transparently.

03 Share

Publish a toolkit another school can understand and reuse.

THEORY OF CHANGE

Students measure
affordable monitors, matched windows

Options compared
costs, maintenance, limits

Effect verified
with outdoor air and a comparison room

School decides
with evidence, not guesses

Others replicate
free toolkit and templates

Objectives are proposed targets for the pilot, not results.

THE AIRSHIFT SOLUTION

Detect → Decide → Verify → Replicate

01

DETECT

Measure PM2.5 and record room conditions, sensor limitations and sampling times.

OUTPUT

Documented baseline

02

DECIDE

Compare source control, portable HEPA filtration and a facilities review, with costs and maintenance.

OUTPUT

Site-specific action plan

03

VERIFY

Measure again under matched conditions. Report change with outdoor air and uncertainty.

OUTPUT

Transparent impact summary

04

REPLICATE

Publish the method, templates, budget variants and lessons learned.

OUTPUT

School implementation toolkit

OBSERVED measured values with dates and windows

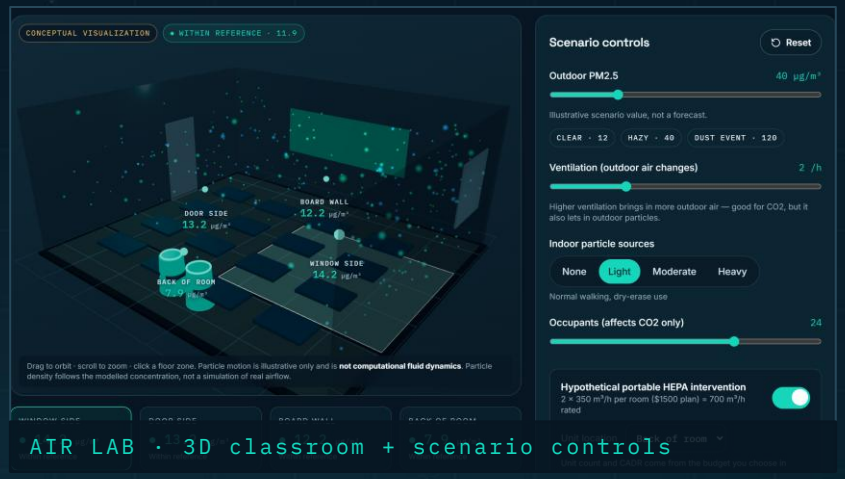
ESTIMATED modelled ranges with stated assumptions

HYPOTHETICAL illustrative scenarios, never results

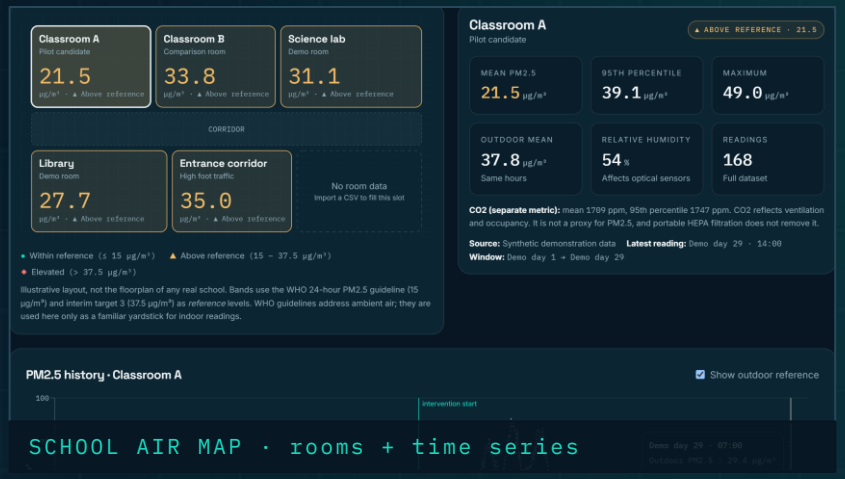
Intervention guidance: US EPA, Guide to Air Cleaners in the Home; US CDC/NIOSH, Ventilation in Schools and Childcare Programs.

INTERACTIVE PROTOTYPE

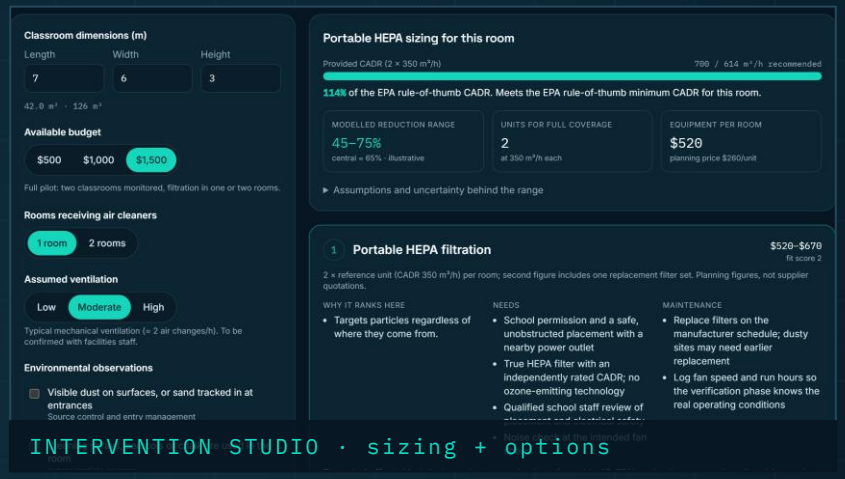
A working decision tool, not a mockup.



AIR LAB · 3D classroom + scenario controls



SCHOOL AIR MAP · rooms + time series



INTERVENTION STUDIO · sizing + options



EVIDENCE LAB · before/after + warnings

SIX MODULES

- Cinematic landing page
- Air Lab (3D)
- School Air Map + CSV import
- Intervention Studio ($500 / $1,000 / $1,500)
- Evidence Lab + report export
- Global Replication + toolkit

BUILT WITH

React, TypeScript, Vite, Three.js, Recharts. Offline, no API keys. Keyboard accessible, reduced-motion aware, responsive.

LIVE DEMO
[URL to be added after publishing]

Screenshots are from the built application. All readings shown are synthetic demonstration data. Particle animation is conceptual, not computational fluid dynamics.

PROJECT STRUCTURE

One student lead. A school community. Eight weeks.

MAIN GOALS

- Establish a documented PM2.5 baseline for one or two classrooms
- Choose and test one affordable intervention, only if the baseline justifies it
- Verify the effect transparently, with limits
- Publish a reusable school toolkit

KEY ACTIVITIES

- Request school permission and agree the protocol
- Measure school-hour PM2.5 with matched windows
- Record sources, ventilation and cleaning routines
- Implement the approved intervention with staff review
- Analyze, report and share with the community

ROLES & RESOURCES

- Lead (solo): research, tool, measurement, analysis, education, reporting, outreach
- School staff: review placement and infrastructure concerns
- Equipment from the grant (see funding)
- About 8 weeks of student time outside class
- Free open-source software

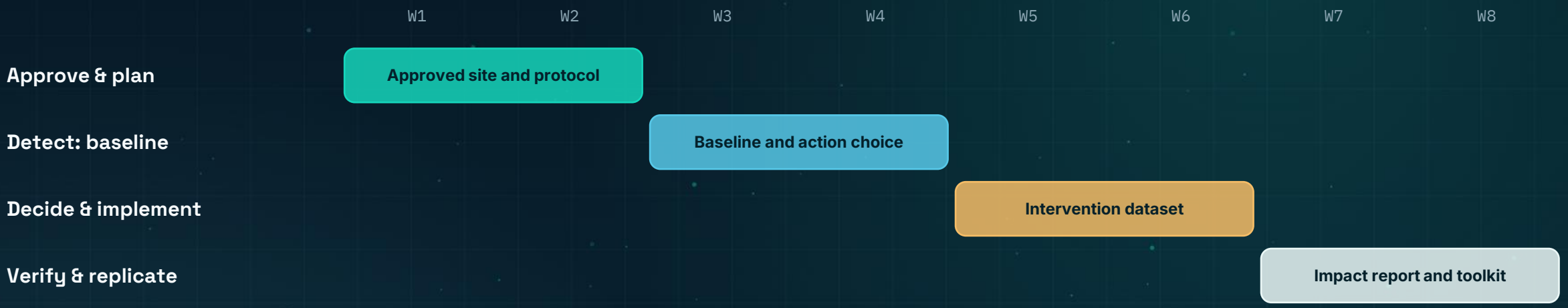
POTENTIAL STAKEHOLDERS · TO BE ASKED, NONE CONFIRMED

SCHOOL LEADERSHIP | FACILITIES & HVAC STAFF | TEACHERS OF THE ROOMS | CLEANING STAFF | STUDENT PEERS (OUTREACH)

Stakeholders listed are people who would need to be asked; none has agreed yet. No student-identifiable data is collected.

IMPLEMENTATION ROADMAP

Eight weeks from funding to published toolkit.



Dependencies: school approval → baseline review → intervention (only if justified) → matched follow-up → report

CHALLENGE	MITIGATION
School approval is delayed or declined	Ask in week 1; fall back to one teacher-approved classroom or a space the school nominates.
Sensors fail or a dust event dominates a window	Two monitors, side-by-side check, spare days in schedule. Record the event rather than discarding it.
Baseline is already within guideline levels	Report it as a finding. Do not buy equipment that is not justified; use funds for outreach and toolkit.
Results are inconclusive or confounded	Publish with caveats; use outdoor air, humidity and a comparison room. Inconclusive evidence still helps the next school.

Planned activities and proposed targets. Dates start when funding is confirmed. Planning prices will be replaced by supplier quotations before any purchase.

FUNDING AND IMPACT

$1,500 buys a complete, honest pilot.

FULL REQUEST · $1,500



Particulate monitoring (2 devices)	$360
Portable HEPA filtration	$520
Replacement filters & maintenance	$150
Data-logging & technical supplies	$110
Comparative measurement supplies	$100
Education & outreach	$90
Contingency	$170
Total	**$1,500**

$1,000

Focused pilot

One treated and one comparison room. Two lower-cost monitors, two air cleaners. Thin contingency, lighter outreach.

$500

Single-room demo

One monitor, one smaller air cleaner. No simultaneous comparison; coverage of recommended airflow is partial.

PROPOSED SUCCESS INDICATORS (TARGETS)

- ≥ 10 school days of baseline and of follow-up per room, ≥ 90% complete
- PM2.5 summaries in µg/m³ with dates, windows and sensor caveats
- Equipment run-hours, maintenance and cost per classroom recorded
- Toolkit read-through by ≥ 3 outside educators or student groups

EVALUATION PROTOCOL

Same sensors, placement and school hours · outdoor air alongside · comparison room · transition day excluded · uncertainty and data-quality warnings reported · no health claims.

Preliminary planning figures, not supplier quotations; to be checked against local quotes, room size and school approvals before purchase. Indicators are proposed targets, not results.

GLOBAL VISION AND CLOSING

From one classroom to a method others can copy.



PUBLISH	Method, templates and budget variants, free to use.
TEST	Can outside educators follow it without help?
SUPPORT	Peer schools repeat it with their own approved equipment.

THE ASK
The $1,500 first-place award funds the full pilot. $1,000 or $500 funds a smaller, equally honest version.

You can’t fix what you can’t see.

AIRSHIFT makes the air visible, measures whether the fix worked, and shares it.

Globe markers other than Al Dhafra are hypothetical regions for illustration only. No partners, sites or deployments exist outside the proposed pilot.